

Answers without supporting work will not be given full credit, so make sure to explain the theoretical basis of what you do for full score.

No calculators/computers or books/notes are allowed.

Write your name and student number in all the papers you submit. Papers without name will not be graded.

Do *not* submit your scrap paper, only the paper where the final answers and explanations are written.

Do *not* write your answers with pencil or red/green pen. Only blue/black pen.

Grade = # points + 1

Question 1 (1 pts) Compute the following limits:

a) (0.5 pts) $\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1}$

b) (0.5 pts) $\lim_{x \rightarrow \infty} \frac{-3x^4 + 5x^2}{x^3 - x + 7}$

Question 2 (1.5 pts) Let $f(x) = \frac{\sin x}{x}$. Find $\lim_{x \rightarrow 0} f(x)$.

Hint: When x is close to 0, the function satisfies that $\cos x \leq f(x) \leq 1$.

Question 3 (1.5 pts) Show that there exists a solution to the equation $x + \sin x = 1$.

Hint: Use Bolzano's Theorem.

Question 4 (1.5 pts) The critical points of a function f are those x for which $f'(x) = 0$. Find the critical points of $f(x) = \sqrt[3]{x^2}(2x - 1)$.

Question 5 (2.5 pts) Calculate $\int_0^1 \log x \, dx$, where $\log x$ is the natural logarithm of x .

Hint: The function $g(x) = x$ goes to zero faster than $f(x) = \log x$ goes to $-\infty$ close to $x = 0$.

Question 6 (1 pts) Compute the Jacobian at the origin of the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $f(x, y, z) = (e^x e^y, y - x^2 z, z \cos y + x^3)$.